

INTELLIGENT RESUME SCREENING SYSTEM

Using Machine Learning and Natural Language Processing

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Project Type: Major Internship Project

Domain: Artificial Intelligence / Machine Learning

Platform: Google Colab

Submission Deadline: 13 August 2026

Problem Statement

Recruiters spend excessive time manually screening resumes. When a large number of applications are received, manually comparing candidate qualifications with job requirements becomes time-consuming and inefficient. An automated system is required to analyze resume text, compare it with a job description, calculate relevance, and rank candidates according to suitability.

Project Overview

The Intelligent Resume Screening System is an AI-driven recruitment automation project that ranks resumes based on their relevance to a given job description. The system processes resume text, removes unnecessary content, performs Natural Language Processing (NLP), converts text into numerical features using TF-IDF, and calculates similarity between a job description and each resume using cosine similarity. The resulting relevance scores are used to produce a ranked list of candidates.

Objectives

- Load and inspect the resume dataset.
- Extract readable text from HTML-formatted resumes.
- Preprocess resume text using NLTK.
- Remove unnecessary characters and stopwords and perform lemmatization.
- Convert resume text into numerical features using TF-IDF.
- Compare resumes with a supplied job description using cosine similarity.
- Calculate relevance and match percentages.
- Rank resumes according to their suitability.
- Provide a foundation for automated recruitment screening.

Dataset Information

Dataset Name: Resume Dataset

Source: Kaggle

Link: <https://www.kaggle.com/datasets/snehaanbhawal/resume-dataset>

The dataset contains resume information and category labels. The project uses resume HTML/text content as the main input for text processing and matching.

Tools & Technologies Used

- Python
- Google Colab
- Pandas
- NumPy
- NLTK
- Scikit-learn
- BeautifulSoup
- Matplotlib / Seaborn
- TF-IDF Vectorization
- Cosine Similarity

System Workflow

Resume Dataset → Text Extraction → Text Preprocessing → TF-IDF Feature Extraction → Job Description Input → Cosine Similarity → Relevance Score → Resume Ranking → Top Candidates

Methodology

1. Data Loading

The CSV resume dataset is loaded into a Pandas DataFrame in Google Colab.

2. Text Extraction

The dataset contains HTML-formatted resume content. BeautifulSoup is used to remove HTML tags and extract readable resume text.

3. Text Preprocessing

NLTK is used to convert text to lowercase, remove special characters and numbers, remove English stopwords, and perform lemmatization.

4. Feature Extraction

TF-IDF (Term Frequency–Inverse Document Frequency) converts cleaned resume text into numerical vectors.

5. Relevance Calculation

The job description is processed using the same pipeline and transformed into a TF-IDF vector. Cosine similarity is calculated between the job-description vector and each resume vector.

6. Ranking

Similarity values are converted into match percentages. Resumes are sorted from highest to lowest score to identify the most relevant candidates.

AI Techniques Used

- Natural Language Processing (NLP)
- Text preprocessing
- Feature extraction using TF-IDF
- Cosine similarity-based relevance scoring
- Automated candidate ranking

Implementation / Work Evidence

Insert actual screenshots from the completed Google Colab notebook in this section:

- Library installation and imports.
- Dataset upload and DataFrame preview.
- Dataset columns and missing-value check.
- HTML-to-text extraction output.
- NLTK preprocessing output.
- TF-IDF vectorization output.
- Job description input.
- Cosine similarity and ranked candidate output.
- Top candidate table with match percentages.
- Top-match visualization.
- Exported CSV results.

Expected Output

The system should generate a ranked candidate table containing rank, resume/category, match percentage, and suitability level. Exact scores depend on the job description and resumes processed.

Rank	Resume / Category	Match %	Suitability
1	Top matching resume	Generated by system	Highly Suitable
2	Second matching resume	Generated by system	Suitable
3	Third matching resume	Generated by system	Suitable

Screenshots / Work Evidence

Add the final Google Colab screenshots here. Evidence should cover the dataset, preprocessing, TF-IDF, similarity results, ranking, and visualization.

Conclusion

The Intelligent Resume Screening System demonstrates how Natural Language Processing and machine learning techniques can automate an important part of recruitment. By transforming resume text into TF-IDF features and comparing it with a job description using cosine similarity, the system ranks resumes according to textual relevance. The project provides practical experience in NLP, feature extraction, similarity-based machine learning, data processing, and recruitment automation.

Future Scope

- Automatic skill extraction from resumes and job descriptions.
- Education, experience, certification, and skill-based scoring.
- Comparison of multiple classification approaches.
- Recruiter dashboard for interactive screening.
- Bulk resume uploads.
- Downloadable candidate reports.
- Flask-based web deployment.
- Fairness and transparency improvements in automated ranking.

External Link

GitHub Repository: <https://github.com/Eshaan1234/Major-Project-Internship-Intelligent-Resume-Screening-System>

Dataset: <https://www.kaggle.com/datasets/snehaanbhawal/resume-dataset>

References

Kaggle Resume Dataset.

NLTK documentation.

Scikit-learn documentation for TF-IDF and cosine similarity.